ATTACHEMENT A

Statement of Work

MICROPLASTICS ANALYSES

Analysis requires a laboratory equipped with an Agilent 8700 Laser Direct Infrared (LDIR) Chemical Imaging System. This system uses quantum cascade laser (QCL) imaging technology which is a new method for spectral analysis and chemical imaging that advances the ability to detect microplastic (MP) particles, determine their size, and acquire detailed information about their chemical composition. Compared to conventional spectroscopic or microscopic methods, LDIR enables efficient and accurate quantification and identification of MPs in sizes between 10 µm and 500 µm.

QUALITY ASSURANCE/QUALITY CONTROL

To address potential interferences by non-plastic particles from the samples, a software MP library should exist that contains various spectral data of non-microplastics (e.g., rubber, naturally occurring polyamide, chitin, cellulosic materials, carbonates, and stearates which are commonly present in environmental samples).

For field data collection, a field blank should be collected using deionized water to monitor for airborne and procedural contamination. The field blank should be prepared by rinsing clean sampling equipment and collecting the rinse water for analysis. The laboratory should analyze procedural blanks when filtering samples. Blank samples should have the same preservation methods and holding times as that of an environmental sample.

DATA REPORTING

A MP analytical report will be provided by the laboratory that contains methodology description, quality control summary, table of total number of MP detected, chart of MP size distribution, results of the method blank, laboratory control sample, sample summary, and method summary.

SAMPLING HANDLING

Sample Storage:

- Place glass bottles in a cooler box with ice packs (preferred method) at the field site to minimize biological activity. However, if an ice cooler box is not available, it is acceptable to transfer the samples directly into a refrigerator maintained at approximately 4°C upon returning to the laboratory.
- Avoid freezing to prevent MP structural damage.
- Store samples in the dark to avoid UV degradation.

Sample Preservation and Holding Time

- Preservation Method:
  - Store samples in a cool (approximately 4°C) and dark environment after collection.
  - Avoid exposure to direct sunlight or heat to prevent degradation or biofouling.
  - Do not freeze water samples, as freezing and thawing may alter the physical characteristics of microplastics.
- Holding Time:
  - Analyze samples within 7-14 days of collection for best integrity.
  - If storage exceeds 14 days, validate through quality control whether degradation or changes have occurred.
  - For long-term storage, samples may be filtered immediately in the lab and the filters dried and stored until analysis.

GENERAL REQUIREMENTS

The services provided by this order are Non-Personal Services. The contractor shall furnish all facilities, labor, and materials to provide goods/services in accordance with the terms and conditions herein and the specifications set forth above.